

Combating Online Misinformation Videos: Characterization, Detection, and Prevention

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Abstract

Recent progress of generative AI and the popularity of short-form video-sharing platforms have raised new risks of misinformation video issues, posing a potential threat to online multimedia ecosystems. With the aid of generative AI tools, producing and spreading vivid, persuasive misinformation videos has been easier, while detecting and preventing them has become harder. This tutorial introduces how to characterize, detect, and prevent misinformation videos, which consists of three technical parts: 1) Characterization of AI-generated and human-edited misinformation videos; 2) Detection approaches, covering those tailored for fully generated, manipulated, and human-edited videos; and 3) Prevention strategies, including those effective for the creation and spread phases. This tutorial concludes by discussing the status quo and ongoing challenges and highlighting the promising directions for future research. We expect to bring broader attention to misinformation video issues, gather and communicate with researchers of interest, and facilitate the engagement of those who are new to this field.

CCS Concepts

• Information systems → Multimedia information systems.

Keywords

Misinformation Video, Fake News Detection, Video Generation

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1 Motivation and Implication

Due to the increasing popularity of short-form video-sharing platforms like TikTok and YouTube Shorts, watching videos to get information has been common for multimedia consumers. It is reported that over 50% of U.S. adults use TikTok to regularly get news¹. Moreover, the surge of generative AI techniques has reshaped the video production pipeline. Powerful generation models like Sora have demonstrated great capabilities for high-definition, vivid videos, making video production easier and more economical than before. Unfortunately, induced by the two key changes, misinformation videos have newly emerged and posed serious threats in reality. Misinformation creators can produce a large number of misinformation videos that realize the underlying malicious aims. For example, an Internet ghostwriter gang created 2.8k videos and earned over 2.7B views². Therefore, it is of urgent need to pay attention to misinformation video moderation and secure the health of the online multimedia ecosystem [1]. In this tutorial, we focus on three levels of misinformation video issues, i.e., characterization, detection, and prevention. This half-day tutorial is open to those with a basic knowledge of multimedia forensics and generative AI. The tutorial will be both *introductory* for those who are of broad interest in AI-generated content safety and *advanced* for those who research multimodal misinformation/deepfake moderation. Attendees will have access to resources via <https://misinfo-video.github.io/>.

2 Course Description

1. Introduction: We start by showing the change in online information consumption and the cases of generative AI in misinformation production, introducing the motivation and the scope.

2. Preliminaries: Misinformation videos are usually either human-edited or AI-generated. We first introduce the basics of video editing operations and show common ways like misleading material cutting and out-of-context text posting to be used in misinformation video creation. Then we introduce the essence of typical video generation models, including a brief technical history and the applications.

¹<https://www.pewresearch.org/short-reads/2024/09/17/more-americans-regularly-get-news-on-tiktok-especially-young-adults/>

²<https://baijiahao.baidu.com/s?id=1813935138703500389>

3. Characterization: This part first overviews the available datasets by introducing the construction pipeline, key statistical indicators, and cases. The characteristics of AI-generated videos include the analysis result from the perspectives of visual artifacts [7], frequency abnormalities [15], temporal consistency [3], and fingerprints of different generation models [17]. The characteristics of human-edited videos include the statistical results from the perspectives of text, audio, video, social context, and temporal propagation patterns [2, 11]. This part aims to shape an intuitive perception of misinformation videos for audiences.

4. Detection Part I (AI-generated misinformation): Detection techniques of AI-generated misinformation videos capture generation traces left by errors or generators' fingerprints. Existing approaches take forensics signals like noise features [10], frequency features [6], and frame-level abnormalities [14] as indicators. We also include a branch of methods that improves explainability [12]. The more fine-grained model attribution predicts the source generator of a given generated video, including model-level, architecture-level, and open-set attribution [17, 18].

5. Detection Part II (human-edited misinformation): The detection techniques of human-edited misinformation videos exploit signal-, semantic-, and intent-level clues. Signal-based methods mine editing traces left by multimedia editing software, including active and passive methods. Semantic-level methods include leveraging companion texts, extracting inter-frame visual features, and checking acoustic features, in addition to cross-modal correlation modeling methods [11]. Intent-based methods model social contexts to reflect the creators' intentions [8]. This part will be finalized with clue integration techniques including parallel and sequential integration to construct the detection pipeline.

6. Prevention: The prevention strategies apply in both the creation and spreading phases. Techniques for creation prevention include embedding tamper-proof digital identifiers [5], tracing the context of multimodal materials [16], and mitigating hallucinations in multimodal large models [19]. Key perspectives for spread prevention include authenticity alerts and psychological inoculation, intervening in the spread of misinformation based on recommendation systems [13], and promoting the truth through interactive and personalized debunking [9].

7. Conclusion and Discussion: We conclude by recapping the key points, listing unsolved challenges like generalization, explainability, and efficiency, and raising open questions like how to measure and audit the real-world effects [4] and how detection and prevention cooperate.

3 Brief Outline

- (1) **Introduction** (10'): Background, motivation, scope, and tutorial overview
- (2) **Preliminaries** (15'): Video editing & generation and types of misinformation videos
- (3) **Characterization** (20'): Dataset overview and characteristics of AI-generated and human-edited misinformation videos
- (4) **Detection Part I: AI-generated misinformation** (40'): Generated and manipulated video detection, and video generation model attribution

- (5) **Detection Part II: Human-edited misinformation** (40'): Signal-based, semantic-based, and intent-based detection, and clue integration for misinformation video detection
- (6) **Prevention** (40'): Prevention of creation and spread
- (7) **Conclusion and Discussion** (15'): Summary of key points; emerging trends and challenges; and open questions, research directions, and resources

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